Cool summers are stormy summers: precession reorganizes extratropical cyclone activity through the quiet season

Abstract

Extratropical cyclones cluster in winter, while precession reshapes the seasonal insolation cycle most strongly in summer, so storm tracks might be expected to ignore the precession cycle. This expectation is tested with twelve equilibrium simulations of the coupled climate model AWI-ESM-2.1 that cover the precession cycle at intervals of thirty degrees under amplified eccentricity and fixed obliquity, with extratropical cyclones tracked explicitly in six-hourly pressure fields in both hemispheres. The annual-mean storm response is small. The seasonal response is not. In the North Atlantic, the North Pacific, and the Southern Ocean the response concentrates in local summer, where storm activity ranges across the phases by up to two thirds of its mean while winter changes stay below about a tenth. We find that summers spent near perihelion are calm and summers spent near aphelion are stormy, so midlatitude storminess in the two hemispheres varies in antiphase across the precession cycle. The signal is carried by cyclone number and position rather than by propagation speed, and it follows the pronounced summer response of the Eady growth rate, which is dominated by vertical shear in all three basins. Precession therefore acts on extratropical storm tracks through their quiet season. Summer-sensitive proxy archives should record a strong and hemispherically antiphased storminess signal at the precession period that annually integrating archives would miss.

1 Introduction

Extratropical cyclones organize the weather of the middle latitudes. They carry most of the poleward heat and moisture transport outside the tropics, deliver a large share of midlatitude precipitation, and cluster into the storm tracks of the North Atlantic, the North Pacific, and the Southern Ocean [Hoskins and Valdes, 1990, Chang et al., 2002, Hoskins and Hodges, 2002]. These storms grow by baroclinic instability on the meridional temperature gradient [Eady, 1949], and the gradient is steepest in winter. Storm activity therefore peaks in the cold season in both hemispheres, in the south as clearly as in the north [Simmonds and Keay, 2000, Hoskins and Hodges, 2005]. Extratropical cyclones are creatures of winter.

Precession, in contrast, is a lever that acts on summer. The slow turning of the longitude of perihelion is the strongest orbital pacemaker of the seasonal insolation cycle, yet it redistributes sunlight rather than adding any, and the redistribution is proportional to the sunlight a season already receives [Berger, 1978, Laskar et al., 2004]. Midlatitude summer insolation swings by more than 100 W m^{-2} across a precession cycle while midwinter insolation barely moves, which is why the orbital literature treats summer energy as the climatically active quantity [Huybers, 2006]. A winter phenomenon facing a summer lever invites a simple expectation. Precession should have little to say about storm tracks. That expectation has never been put to a clean test.

Where the imprint of precession has been studied in depth, it appears as a seasonal reorganization of climate rather than a change in the annual mean. The monsoon literature carries this tradition, from the classic orbital experiments of Kutzbach and Guetter [1986] to single-forcing designs that isolate precession explicitly [Bosmans et al., 2015, Merlis et al., 2013], and cave records show the northern and southern monsoons swinging in antiphase as perihelion migrates through the year [Wang et al., 2008]. The extratropics have received far less attention. Kaspar et al. [2007] simulated a stronger and poleward-shifted North Atlantic winter storm track for the Eemian from two orbital time slices, Brayshaw et al. [2010] traced Holocene orbital forcing into the North Atlantic and Mediterranean winter storm tracks through the baroclinicity they support, and Varma et al. [2012] found the Southern Hemisphere westerlies migrating under Holocene orbital forcing. Explicit cyclone tracking has entered paleoclimate work mainly at the Last Glacial Maximum, where ice sheets dominate the forcing [Pinto and Ludwig, 2020, Raible et al., 2021], and in transient simulations of the last millennium [Hess and Chemke, 2025]. For tropical cyclones the precessional response has recently been isolated in high-resolution paleoclimate simulations [Raavi et al., 2023]. To our knowledge, no previous study has applied explicit cyclone tracking across a full precession cycle to isolate the orbital response of the extratropical storm tracks.

Closing this gap matters beyond dynamics. However, the paleoclimate records that would register past storminess, from dust and wind-blown sediment to westerly-belt precipitation, respond to particular seasons rather than to the annual mean, and a seasonally biased archive can fabricate or erase an orbital signal depending on which season carries it [Laepple et al., 2011]. Knowing where in the seasonal cycle the precessional storm signal lives is a prerequisite for reading such records. Storm diagnostics raise a parallel issue, because measures of cyclone influence time, cyclone counts, and cyclone intensity respond to forcing in different ways and must be separated before any single proxy is interpreted.

In the present study the precession cycle is covered with twelve equilibrium simulations of a coupled climate model, and every extratropical cyclone is tracked explicitly in six-hourly pressure fields in both hemispheres. The design isolates precession at amplified eccentricity with all other boundary conditions fixed, so that the seasonal anatomy of the orbital storm response can be mapped basin by basin against the calendar month of perihelion.

The naive expectation fails, and it fails in an instructive way. The annual-mean storm response is indeed small, but the signal has not vanished. It is concentrated in local summer, the quiet season, in all three storm basins. Summers spent near perihelion are calm and summers spent near aphelion are stormy, so the two hemispheres respond in antiphase across the precession cycle, an extratropical counterpart of the monsoon seesaw. The response travels from insolation to storms through the summer Eady growth rate, and the growth-rate change is carried by vertical shear [Lindzen and Farrell, 1980]. The paper is organized around which season carries the precessional storm response, what dynamical chain transmits it, and why the winter storm track stays orbitally quiet. Section 2 describes the model, the ensemble, and the tracking. Section 3 presents the seasonal anatomy of the response, and section 4 discusses mechanisms, implications for proxy records, and limitations.

2 Data and methods

2.1 Model and experiments

The simulations are performed with AWI-ESM-2.1-wiso, an isotope-enabled version of the AWI Earth System Model [Shi et al., 2023]. The atmosphere component is ECHAM6 at T63 resolution with 47 vertical levels [Stevens et al., 2013], which includes the land surface module JSBACH [Raddatz et al., 2007]. The ocean and sea ice component is FESOM2, a finite-volume model formulated on an unstructured mesh [Danilov et al., 2017, Sidorenko et al., 2019]. The water isotope tracers carried by the model are not used in the present study.

Twelve equilibrium simulations are analyzed, taken from the idealized precession ensemble introduced by Shi et al. [2025]. The longitude of perihelion ω is prescribed at values from 0° to 330° in steps of 30°, while eccentricity is fixed at 0.058 and obliquity at 24.5°. The eccentricity value sits near the upper bound reached during the Quaternary and amplifies the precessional modulation of insolation, so that the response emerges clearly from internal variability. All remaining boundary conditions follow the preindustrial setup, and each phase is branched from a long preindustrial control. Thirty years of six-hourly output from the equilibrated part of each simulation form the basis of the cyclone analysis. The calendar month of perihelion for each phase is computed from two-body orbital geometry [Berger, 1978], and the same formulary provides the insolation fields shown in Fig. 1.

2.2 Cyclone detection and tracking

Extratropical cyclones are detected and tracked with the algorithm of Crawford et al. [2021]. Six-hourly sea level pressure is first interpolated onto 100-km equal-area grids centered on each pole, and the two hemispheres are processed separately. Cyclone centers are identified as closed low-pressure minima with a pressure gradient of at least 750 Pa relative to the surroundings, and centers closer than 1000 km are merged into multicenter systems. Grid cells with surface elevation above 1500 m are excluded, and the cyclone area is defined by the outermost closed isobar obtained with a 200 Pa contour interval. Centers are linked into tracks across consecutive six-hourly fields subject to a maximum displacement criterion. Only systems poleward of 30° latitude enter the analysis.

Several track-based diagnostics are derived. Track density counts every six-hourly cyclone position, system density counts each cyclone once at its mean position, and genesis density counts the first point of each track. Tracks that merely cross the boundaries of the monthly processing windows would appear as spurious genesis or lysis events, and such truncated events (about 7% of the total) are removed. For every cyclone the lifespan, the maximum depth relative to the outermost closed isobar, the minimum central pressure, the mean cyclone area, and the propagation speed are recorded.

2.3 Gridded storm coverage

A complementary Eulerian diagnostic measures how long each location spends under cyclone influence. At every six-hourly time step a binary mask marks the grid cells that lie within the area of a detected cyclone, and the masks are accumulated into monthly storm days on the T63 grid. Storm days at a point therefore scale with the product of cyclone number and cyclone area footprint. The two factors can carry opposite seasonal cycles, and over the Southern Ocean the larger size of summer cyclones raises summertime storm days even though cyclone counts peak in winter. Frequency statements in this paper consequently rest on the track-based densities, while storm days serve as a smooth field for maps and for measures of cyclone influence time. The gridded product is available for eleven phases, because the mask product of the $\omega = 30°$ experiment stems from an inconsistent processing version and is withheld. All figures use the common set of eleven phases, and the Lagrangian statistics of the withheld phase (which exist for all twelve) confirm that its omission does not affect any conclusion.

2.4 Baroclinicity diagnostics

The maximum Eady growth rate serves as the measure of baroclinicity [Eady, 1949, Lindzen and Farrell, 1980, Hoskins and Valdes, 1990]. It is computed as $\sigma = 0.31\,|f|\,|\partial U/\partial z|\,N^{-1}$ in the 925–700 hPa layer from monthly wind, temperature, and geopotential height. The response to precession is quantified between the end members p090 and p270, which place perihelion at the June and December solstices. Because the difference fields are dipolar, plain box averages cancel, and the response amplitude is instead defined as the root-mean-square of the difference over a basin, normalized by the climatological growth rate of the season. A substitution decomposition in the style of Camargo et al. [2007] splits the difference field into a vertical-shear factor and a static-stability factor, and the share of each factor is obtained by projecting it onto the full difference pattern. The basin averages use the storm-track boxes defined below. This choice matters in the Southern Hemisphere, where a sector beginning at 30°S would admit the seasonal signal of the subtropical winter jet and disguise it as a storm-belt response.

2.5 Basins and statistics

Three storm-track basins are analyzed, the North Atlantic (30–75°N, 80°W–20°E), the North Pacific (30–75°N, 120°E–120°W), and the Southern Ocean (40–75°S, all longitudes). Basin means are area weighted. The size of the precessional response is reported as the range across the eleven phases, the maximum minus the minimum of the 30-year mean, expressed as a percentage of the phase mean. Uncertainty estimates use the interannual spread of the 30 individual years in each phase, and the significance of the end-member difference maps is assessed with a two-sided Welch t test at each grid point. The local-summer contrasts between extreme phases far exceed the interannual noise, with p values below 10^{-12} in all three basins.

3 Results

3.1 Winter storm tracks and what storm days measure

The three storm tracks of the ensemble live in winter, exactly as observed. Seasonal-mean genesis counts peak in DJF over the North Atlantic and the North Pacific and in JJA over the Southern Ocean, and system counts follow the same calendar in every phase. Cyclones also cut deepest in the cold season, with winter maximum depths exceeding their summer values by roughly half in the northern basins. The winter identity of the storm season is therefore common to both hemispheres and to all twelve orbital configurations (Table 1).

The gridded storm-day field tells a partly different story, and the difference is instructive. Basin-mean storm days over the Southern Ocean stay high through austral summer (Fig. 2c) although summer cyclones are fewer. The elevation reflects cyclone size rather than cyclone number. Summer systems there are about twenty percent larger in area than their winter counterparts, and storm days count the time a grid cell spends inside any cyclone, so the product of number and footprint hides the winter peak in counts. A diagnostic of cyclone influence time is not a diagnostic of cyclone frequency, and the two must be read together. Frequency statements below therefore rest on the track-based densities, while storm days provide the smooth fields used for maps.

3.2 The summer paradox

Precession barely touches the annual mean but reorganizes the seasons, and the season it selects is the quiet one. The seasonal cycles of basin-mean storm days fan out across the phases almost exclusively in local summer (Fig. 2a–c). The across-phase range makes this quantitative (Fig. 2d–f). In the North Pacific the annual range is 17% of the phase mean while the JJA range reaches 67%. The North Atlantic behaves the same way with 10% against 31%. The Southern Ocean repeats the pattern half a year out of phase, with an annual range of only 4% against 16% in DJF. The track-based densities confirm the calendar in every basin, with local-summer ranges of 24%, 43%, and 22% for track density in the North Atlantic, the North Pacific, and the Southern Ocean. The sensitive season is local summer everywhere, and the small annual signal is what survives of a substantial summer swing after the stable winter is averaged in.

3.3 Orbital anchoring and the interhemispheric mirror

The summer response is anchored to the orbit, not to the calendar. When the basin-mean storm-day anomaly is drawn as a function of calendar month and precession phase, the anomaly ridge climbs through the diagram along the calendar month of perihelion (Fig. 3). The sign is fixed by the orbital geometry. Summers spent near perihelion are quiet and summers spent near aphelion are stormy. North Pacific summer activity reaches its minimum at $\omega = 90°$, the configuration with perihelion at the June solstice, where it falls 32% below the phase mean, and its maximum of +35% near $\omega = 270°$ (Fig. 6a). The North Atlantic swings between −17% and +15% at the same phases. The Southern Ocean mirrors this behavior, with a DJF maximum at $\omega = 60°$ and a minimum at $\omega = 210°$, because a perihelion that falls in northern summer is an aphelion summer for the south. One lever thus drives the two hemispheres in antiphase across the precession cycle. The extreme-phase contrasts far exceed interannual variability, with Welch p values below 10^{-12} in all three basins.

3.4 The dynamical bridge runs through summer shear

The seasonal selectivity begins in the forcing itself. The monthly insolation range across the phases at 45°N climbs from below 20 W m^{-2} in midwinter to about 128 W m^{-2} in early summer, and the mirror curve at 55°S peaks near 134 W m^{-2} in December and January (Fig. 1). Perihelion can only amplify a season that already receives sunlight, so the orbital lever presses on summer in each hemisphere and on almost nothing else.

Baroclinicity translates the pressure on the seasons into the storm response. The Eady growth rate difference between the end members p090 and p270 is largest in local summer within every storm basin (Fig. 4). Relative to the seasonal climatology the summer response reaches 31% in the North Atlantic, a striking 92% in the North Pacific, and 21% in the Southern Ocean, in each case the seasonal maximum for that basin. A substitution decomposition attributes the pattern almost entirely to the vertical-shear factor, whose projection accounts for 103–109% of the difference field in the three basins, while the static-stability factor contributes near zero. The chain is short and uniform. Perihelion summers warm the summer hemisphere, flatten the meridional temperature gradient, weaken the thermal-wind shear, and starve the baroclinic growth on which summer cyclones depend. The storm anomalies appear somewhat poleward of the growth-rate anomalies, consistent with the downstream development of baroclinic waves [Chang et al., 2002], and this offset is why pattern measures rather than same-latitude box averages are used throughout. Winter tells a complementary story. The winter growth rate still shifts by 16–18% between the end members, yet the winter storm response stays below 12% in every basin, a point taken up in the discussion.

3.5 Spatial expression, and what precession actually moves

The end-member difference maps show where the summer signal lives (Fig. 5). In the annual mean the p090 minus p270 difference is weak and patchy. In JJA a pronounced dipole spans the northern basins, with fewer storm days through the midlatitude core and more along the Arctic flank at perihelion summer, the signature of a poleward shift joined to an overall summer weakening. In DJF the significant signal migrates to the Southern Ocean and forms a circumpolar banded structure of opposite sign. About half of the JJA grid points in the northern extratropics pass the 95% significance test, against a far smaller fraction in the annual mean.

What precession moves is above all the number and location of cyclones, not their translation speed, and only regionally their depth (Fig. 6b). Local-summer ranges of the frequency and position metrics span 16–67% across the basins. Propagation speed stays within 2–5% everywhere. Cyclone depth separates the basins, and the diagnostics are deliberately kept apart in the manner of Zappa et al. [2013]. Over the Southern Ocean the maximum depth varies by only 4% across the phases while frequency varies by 22%, a clean decoupling of frequency from intensity. Over the North Pacific the summer depth range reaches 33% and the lifespan range 23%, so there the whole cyclone population weakens together when perihelion falls in summer, fewer systems that are also shallower and shorter lived. The North Atlantic sits between these regimes. The common element across basins is that occurrence responds most, while the intensity response ranges from absent to substantial depending on how strongly the summer baroclinic reservoir is drained.

4 Discussion

4.1 Why the lever lands in summer

The seasonal selectivity of the storm response is set by orbital geometry before any model physics enters. Precession redistributes insolation in proportion to the insolation a season already has, so the winter hemisphere offers the lever almost nothing to press on [Berger, 1978]. The monthly forcing amplitude in Fig. 1 makes the floor explicit, with midwinter values near 20 W m^{-2} against summer values above 125 W m^{-2} at the latitudes of the storm tracks. This part of the result is model independent and would reappear in any climate model, because it restates the geometry that also underlies the summer-energy view of orbital forcing [Huybers, 2006]. The novelty is not that summer receives the forcing, which the monsoon literature established long ago [Kutzbach and Guetter, 1986, Bosmans et al., 2015], but that the summer forcing finds a winter phenomenon and reorganizes it through its quiet season.

4.2 Cool summers are stormy summers

The mechanism that converts summer insolation into summer storms is the thermal wind. Perihelion summers warm the summer hemisphere most strongly over land and at high latitudes, flatten the meridional temperature gradient, and weaken the vertical shear that feeds baroclinic growth. The factor decomposition confirms that the shear term carries essentially the whole Eady growth rate response in all three basins, while static stability contributes almost nothing. The same chain, with greenhouse warming in place of perihelion, is now observed in the present-day Northern Hemisphere, where amplified summer warming at high latitudes has weakened the summer circulation, the eddy kinetic energy, and the number of strong summer cyclones [Coumou et al., 2015, Chang et al., 2016, Gertler and O'Gorman, 2019]. Projections extend the analogy, with CMIP-class models thinning summer cyclone activity and the associated baroclinicity through the same shear pathway [Lehmann et al., 2014, Priestley and Catto, 2022]. The correspondence should be read as a shared mechanism rather than a shared scenario, because the warming trend alters the two hemispheres together whereas precession swings them in antiphase. Under both forcings the rule is the same, and it is the rule in the title. Cool summers are stormy summers.

4.3 The orbital invariance of the winter storm track

The stability of winter deserves as much attention as the mobility of summer. Winter storm activity varies by less than about a tenth across the entire precession cycle in every basin, and this is only partly a matter of weak winter forcing. The winter Eady growth rate still shifts by 16–18% between the end members, yet winter cyclone numbers barely respond. The winter storm track appears saturated in the sense that ample baroclinicity is available in every phase, so that moderate changes in the growth rate do not translate into changes in occurrence. Whatever its ultimate cause, the practical consequence is notable. The delivery of winter precipitation to the midlatitude and subpolar continents, the season and pathway that build snowpack and feed ice sheets, is essentially fixed across orbital configurations of the precession cycle. A transient simulation of the last millennium likewise found cyclone counts nearly insensitive to external forcing [Raible et al., 2018, Hess and Chemke, 2025], and the present result extends that insensitivity to the far larger insolation excursions of the orbital band, for winter alone.

4.4 One lever, three basins, and what the records would see

The three basins express one mechanism with different accents, and the diagnostics must be kept apart to see it. Occurrence responds everywhere, with local-summer ranges of 16–67% in frequency and position metrics, while propagation speed is inert. Intensity separates the basins. The Southern Ocean decouples cleanly, with a 4% depth range against a 22% frequency range, the North Pacific weakens as a whole population in perihelion summers, and the North Atlantic lies between. The pattern echoes the long-standing conclusion from anthropogenic scenarios that cyclone number and cyclone intensity respond to forcing through different processes and need not move together [Bengtsson et al., 2009, Ulbrich et al., 2009, Catto et al., 2019]. The storm-day diagnostic adds its own caution, because cyclone influence time folds cyclone size into cyclone number, and over the Southern Ocean the seasonal cycle of size alone can invert the apparent storm season. Any proxy that integrates storm influence rather than storm counts inherits this ambiguity.

For the paleoclimate record the seasonal anatomy carries a concrete and testable message. An archive with summer-biased sensitivity in the midlatitudes should carry a pronounced precession signal in storminess, while an annually integrating archive should carry almost none, and a winter-biased archive should be quietest of all. The seasonal bias of an archive can therefore fabricate or erase an orbital storm signal exactly as it does for temperature [Laepple et al., 2011]. The interhemispheric mirror sharpens the prediction, because summer-sensitive storminess records from the two hemispheres should vary in antiphase at the precession period, the extratropical counterpart of the monsoon seesaw [Wang et al., 2008]. Records of the southern westerlies already show orbital-band migrations [Lamy et al., 2010, Varma et al., 2012], and glacial-state cyclone modeling has shown how strongly reconstructed storminess depends on which cyclone property a proxy senses [Pinto and Ludwig, 2020]. A paired-hemisphere test at the precession band is within reach of existing archives.

4.5 Limitations

Several limitations bound these conclusions. The results rest on one model, one tracking algorithm, and an idealized orbital configuration with eccentricity fixed near its Quaternary maximum, so the simulated amplitudes represent an upper bound that scales with $e \sin \omega$ rather than a reconstruction of any particular epoch. Obliquity is fixed at 24.5°, and interactions between precession and obliquity or ice volume are outside the design. The simulations are equilibrium slices, so transient effects along the precession cycle are not represented. The T63 atmosphere resolves the storm tracks but not mesoscale substructure, and low-level Eady growth rates in the Southern Hemisphere carry a documented sensitivity

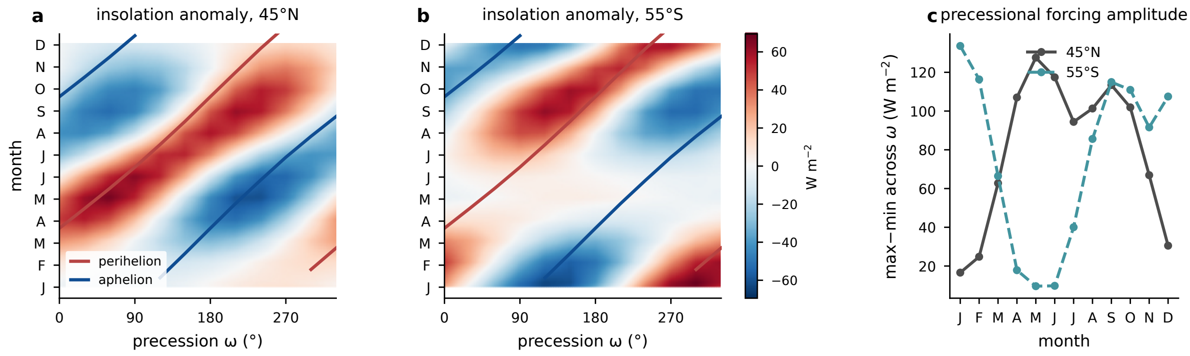

Figure 1: Precessional insolation forcing from orbital geometry with eccentricity 0.058 and obliquity 24.5°. (a) Monthly insolation anomaly relative to the phase mean at 45°N as a function of calendar month and precession phase ω, with the perihelion (red) and aphelion (blue) calendar months of each phase overlaid. (b) As in (a), but for 55°S. (c) Precessional forcing amplitude by calendar month, the maximum minus minimum monthly insolation across the twelve phases, at both latitudes. Units: W m^{-2}.

to boundary-layer stability [Simmonds and Lim, 2009]. The gridded storm-day product uses eleven of the twelve phases, although the complete Lagrangian statistics indicate that the withheld phase would not alter any conclusion.

5 Conclusions

One rule spans the three storm tracks of this precession ensemble. The orbital lever reaches extratropical cyclones through local summer, the quiet season, while the winter storm track stays essentially fixed across the entire cycle. Summers spent near perihelion are calm and summers near aphelion are stormy, and because one hemisphere's perihelion summer is the other's aphelion summer, midlatitude storminess in the two hemispheres swings in antiphase at the precession period. The signal travels through shear-driven summer baroclinicity and expresses itself chiefly in how many cyclones occur and where they run, with per-cyclone intensity responding only where the summer baroclinic reservoir is drained most deeply. For the proxy record the lesson is direct. Orbital storminess signals should be sought in summer-sensitive midlatitude archives, and they should appear with opposite sign in the two hemispheres.

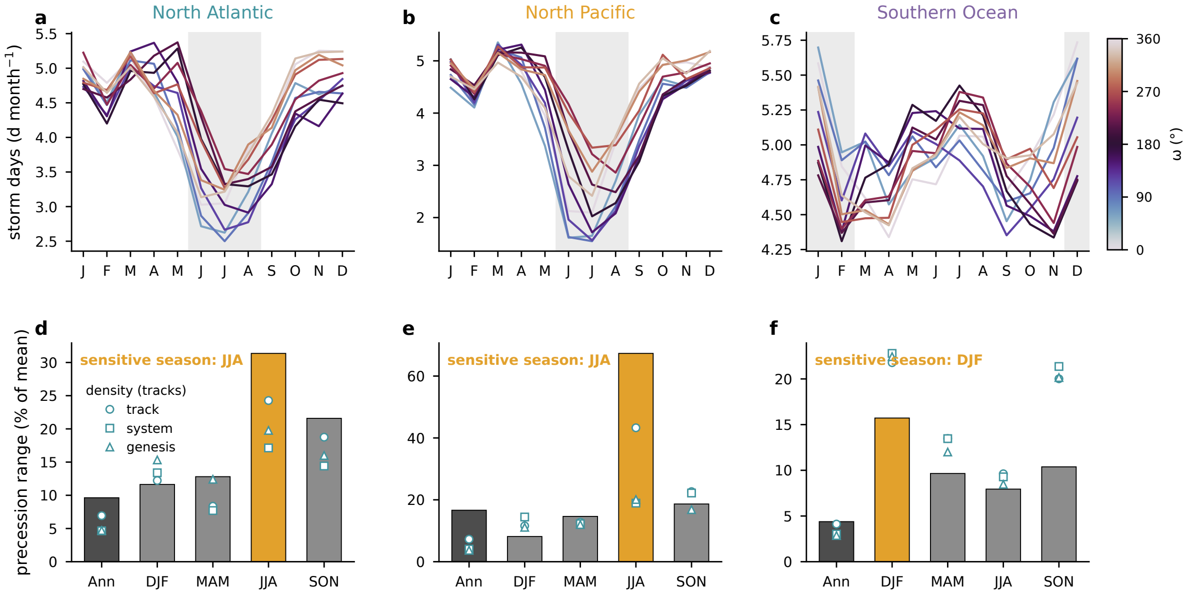

Figure 2: Seasonal concentration of the precessional storm response. (a)–(c) Basin-mean storm days (d month^{-1}) by calendar month for the North Atlantic, the North Pacific, and the Southern Ocean, one line per phase colored by ω, with local summer shaded. (d)–(f) Across-phase range of storm days (maximum minus minimum, percent of the phase mean) for the annual mean and the four seasons, with the local-summer bar highlighted; open symbols give the corresponding ranges of track, system, and genesis densities from the Lagrangian tracks.

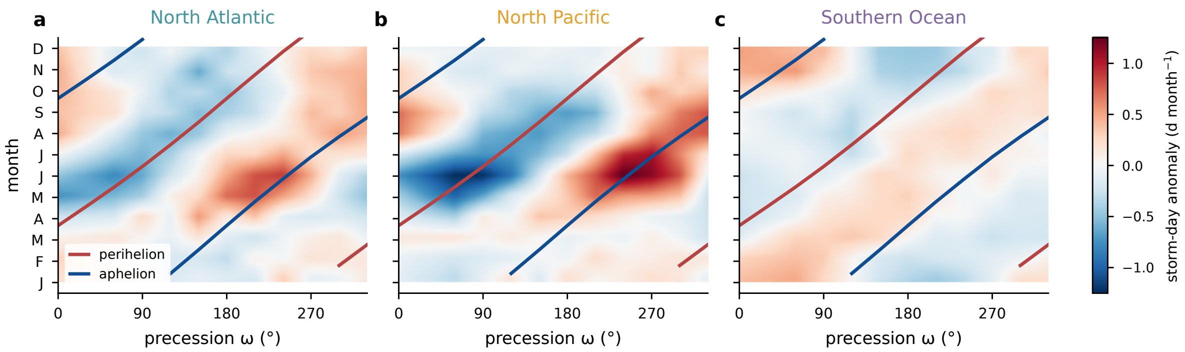

Figure 3: Basin-mean storm-day anomaly (d month^{-1}) relative to the phase mean as a function of calendar month and precession phase for (a) the North Atlantic, (b) the North Pacific, and (c) the Southern Ocean, with the perihelion (red) and aphelion (blue) calendar months of each phase overlaid.

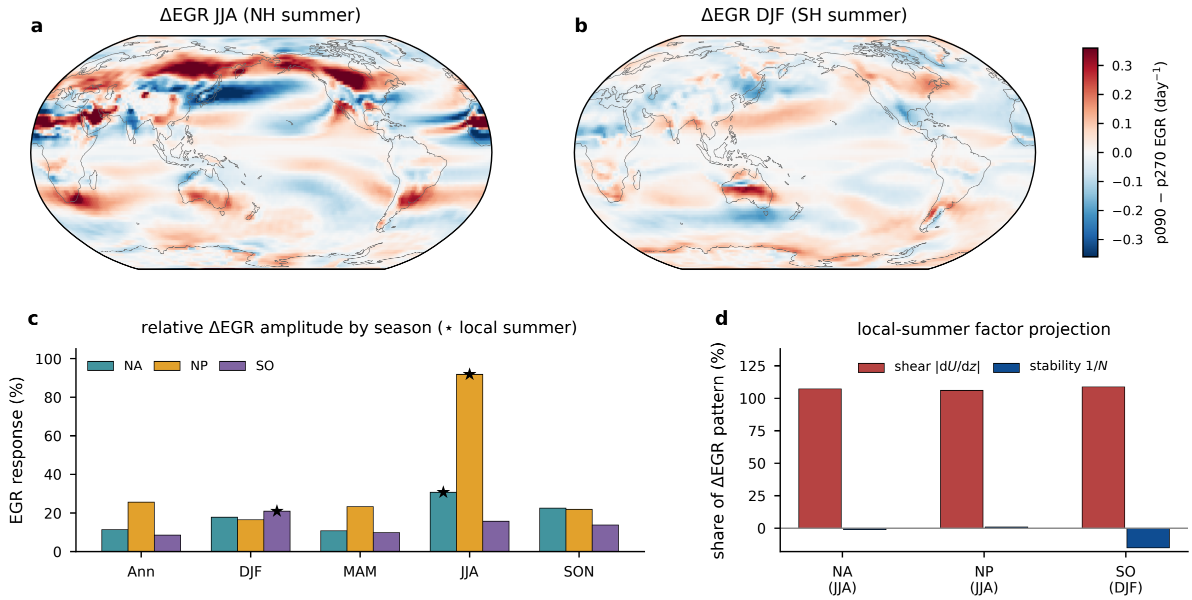

Figure 4: Precessional response of the maximum Eady growth rate (925–700 hPa). (a) p090 minus p270 difference for JJA and (b) for DJF (day^{-1}). (c) Seasonal response amplitude in each basin, the RMS of the p090 minus p270 difference over the storm box as a percentage of the seasonal climatology, with stars marking local summer. (d) Projection shares of the vertical-shear and static-stability factors in the local-summer difference pattern.

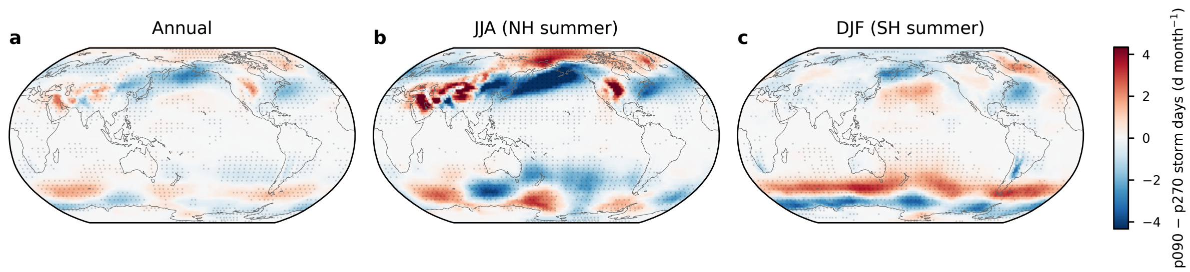

Figure 5: Difference in storm days (d month^{-1}) between p090 and p270 for (a) the annual mean, (b) JJA, and (c) DJF, on a common color scale. Stippling marks grid points where the difference is significant at the 95% level based on a two-sided Welch t test over the 30 simulated years.

Table 1: Three-basin comparison of the precessional storm response. Ranges are max minus min across the 11 phases as a percentage of the phase mean. The storm season is diagnosed from cyclogenesis counts, the sensitive season from the seasonal storm-day range. The EGR response is the RMS p090−p270 difference over the basin sector relative to its climatology; the dominant factor is its projection share.

Basin	Storm season	Sensitive season	Storm-day range ann/summer (heightNorth Atlantic	D
ω=240 / ω=90	31	shear (107%)		
North Pacific	DJF (winter)	JJA	17 / 67	
Southern Ocean	JJA (winter)	DJF	4 / 16	

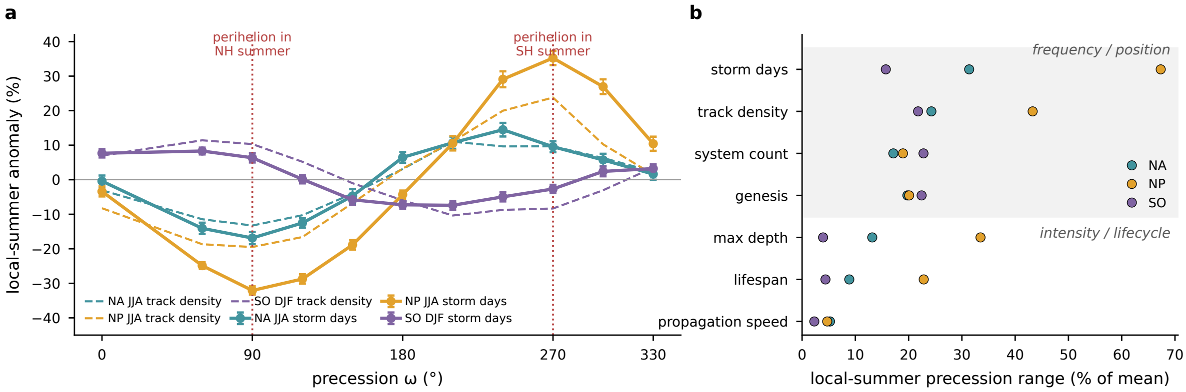

Figure 6: (a) Local-summer storm activity across the precession cycle, shown as the percent anomaly of basin-mean storm days from the phase mean (solid, with interannual standard errors) and of track density (dashed), for the North Atlantic and the North Pacific in JJA and for the Southern Ocean in DJF; dotted vertical lines mark the phases with perihelion in NH and SH summer. (b) Local-summer across-phase range for frequency and position metrics and for intensity and lifecycle metrics in each basin.

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